Ruian Tie

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Education

Fudan University & Shanghai Innovation Institute — *Sep. 2025 – Present*
Ph.D. student. Supervisor: Prof. Xie Chen — Shanghai, China

- Research: singing voice synthesis, streaming music generation.

PLA Space Engineering University & National Meteorological Information Center — *Aug. 2019 – Dec. 2021*
M.S. — Beijing, China

Xidian University — *Aug. 2015 – June 2019*
B.S. — Xi'an, China

Selected Publications

* denotes equal contribution. CCF / CAS tiers are shown where applicable.

[1] ZeroOpera: Zero-Shot Peking Opera Singing Voice Synthesis via Prior-Guided Inpainting and Enhancement
Ruian Tie*, *Jinwen Wang**, *zhengshuang, Haina Zhu, Kang Yin, Jiale Qian, Haohui Xie, Junxi Liu, Chen Ji, ZhuYun, Xie Chen*
Under review at ICLR 2027

[2] Zero-Shot Statistical Downscaling via Diffusion Posterior Sampling
Ruian Tie*, *Wenbo Xiong**, *Zhengyu Shi**, *Xinyu Su, Chenyu Jiang, Libo Wu, Hao Li*
NeurIPS 2026 (CCF-A)

[3] Reconstructing fine-scale 3D wind fields with terrain-informed machine learning
*Chensen Lin**, ***Ruian Tie****, *Shihong Yi, Dongqing Liu, Xiaohui Zhong, Zixin Hu, Hao Li*
Nature Communications 17, 2026 (CAS Q1 Top)

[4] Generative artificial intelligence improves projections of climate extremes
Ruian Tie*, *Xiaohui Zhong**, *Zhengyu Shi**, *Hao Li, Bin Chen, Jun Liu, Libo Wu*
arXiv:2508.16396 · EGU General Assembly 2026 (Highlight Oral)

[5] Improve the downscaling accuracy of high-resolution precipitation field using classification mask
Ruian Tie*, *Chunxiang Shi, Meng Li, Xiang Gu, Lingling Ge, Zhanfei Shen, Jieli Liu, Tao Zhou, Xiaojian Chen*
Atmospheric Research 310: 107607, 2024 (CAS Q1 Top)

[6] To accurately and lightly downscale the temperature field by deep learning
Ruian Tie*, *Chunxiang Shi, Gang Wan, Lihua Kang, Lingling Ge*
Journal of Atmospheric and Oceanic Technology 39(4): 479–490, 2022

[7] CLDASSD: reconstructing fine textures of the temperature field using super-resolution technology
Ruian Tie*, *Chunxiang Shi, Gang Wan, Xingjie Hu, Lihua Kang, Lingling Ge*
Advances in Atmospheric Sciences 39(1): 117–130, 2022 (CAS Q1 Top)

Research Projects

ZeroOpera — zero-shot Peking opera singing voice synthesis — *May 2026 – Present*

- Proposed acoustic inpainting: a Mel-spectrogram from a pop-music base model serves as a structural prior, reconstructed locally with a mask and enhanced globally with opera techniques, preserving the base model's zero-shot timbre generalization.
- Designed a technique-driven phonological modulation module (FiLM) for unseen opera pronunciations, removing the need for fine-grained phoneme alignment.
- Versus SoulX-Singer: 19.4% lower phoneme error rate and +0.43 MOS timbre similarity.

Controllable singing techniques — built on SoulX-Singer — *May 2026 – Present*

- Collected 2M+ raw tracks and built an automatic cleaning pipeline that yields 550K high-quality dry vocals.
- Disentangled timbre from singing technique with GeMAPS features; unsupervised clustering of segment-level features exposes latents for breathy, smoky and belting voice, predicted by an acoustic planner. Word-level control is in development.

Generative priors for spatiotemporal data — downscaling *Mar. 2025 – Apr. 2026*

- Led flow-matching and latent-diffusion frameworks bridging CFD wind fields and multi-model CMIP6 simulations with real observations.
- Proposed latent diffusion posterior sampling with progressive distillation for zero-shot downscaling without retraining.

Experience

Shanghai Academy of AI for Science *Oct. 2024 – Sep. 2025*
Research assistant Shanghai, China

- Generative models for high-dimensional spatiotemporal data; produced publications [2], [3], [4].

National Meteorological Information Center *Jan. 2024 – Sep. 2024*
Research assistant Beijing, China

- Deep-learning downscaling of meteorological fields; published paper [5].

People's Liberation Army *Jan. 2022 – Dec. 2023*
Operations staff officer

Selected Awards & Patents

2nd Prize, 7th All-Army Mathematical Modeling Contest *(2022)*
National 2nd Prize, Huawei Cup China Graduate Mathematical Contest in Modeling *(2020)*
Invention patent CN18520720A: adaptive mesh method for fluid–structure interaction
Software copyrights 2020SR1519123, 2020SR1519122 (epidemic data platforms) *(2020)*

Skills

Languages: Chinese (native), English (CET-6: 451)